Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2024

Code: 110501 | ***Subject: Analog & Digital Communication System*** | *Time: 03 Hours*

Semester: V | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Phase-locked loop (PLL) is used for:
- (i) AM demodulation
- (ii) FM demodulation
- (iii) FM generation
- (iv) None

(b) Which modulation is used in FM radio broadcasts?
- (i) Narrowband FM
- (ii) Wideband FM
- (iii) AM
- (iv) SSB

(c) In a superheterodyne receiver, the purpose of the mixer is to:
- (i) Demodulate the signal
- (ii) Convert frequency
- (iii) Amplify the signal
- (iv) Filter the signal

(d) If the frequency deviation is 75 kHz and modulating frequency is 15 kHz, FM modulation index is.....
- (i) 1
- (ii) 5
- (iii) 2
- (iv) 0.2

(e) Quantization error in PCM can be reduced by:
- (i) Increasing sampling rate
- (ii) Increasing bandwidth
- (iii) Increasing quantization levels
- (iv) Reducing amplitude

(f) In a (7,4) linear block code, how many parity bits are there?
- (i) 4
- (ii) 3
- (iii) 7
- (iv) 1

(g) Which of the following occupies the least bandwidth?
- (i) DSBSC
- (ii) SSB-SC
- (iii) AM
- (iv) VSB-SC

(h) A syndrome is used to:
- (i) Encode the data
- (ii) Transmit the data
- (iii) Detect errors
- (iv) Generate parity

(i) What is the function of a transponder?
- (i) Amplify signals
- (ii) Convert uplink to downlink frequency
- (iii) Modulate signals
- (iv) Code signals

(j) FM bandwidth is calculated using:
- (i) Nyquist formula
- (ii) Shannon’s formula
- (iii) Carson’s Rule
- (iv) Friis equation

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the slope detector method of FM detection with a block diagram.

(b) Draw and explain the basic architecture of a satellite communication system.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Explain the working of a Foster-Seeley discriminator for FM detection.

(b) Explain the working principles of PAM, PWM and PPM with their generation methods.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain the working principle of a square law modulator for generating AM.

(b) Explain the structure and types of optical fibers used in communication.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Define and derive the expression for Signal-to-Quantization Noise Ratio (SQNR).

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(b) Explain DSB-SC RECEIVER using coherent detection.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Consider a delta modulation system designed to accommodate analog message signals limited to band bandwidth of 5 kHz. A sinusoidal test signal of amplitude A = 1 volt and frequency f_m = 1kHz is applied to the system. The sampling rate of the system is 50 kHz.
(i) Calculate the step size Δ required to minimize the slope overload.
(ii) Calculate the signal-to-(quantization) noise ratio of the system for the specified sinusoidal test signal.

(b) Explain the superheterodyne AM receiver with a neat block diagram.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) A sinusoidal modulating signal $m(t) = 2\cos(2\pi \times 10^3 t)$ is used to modulate a carrier $m(t) = 10\cos(2\pi \times 10^5 t)$. The modulation index is 0.4.
(i) Write the expression for the AM signal in the time domain.
(ii) Compute the amplitudes of the carrier and sideband components.
(iii) Calculate the total power of the AM signal if the load resistance is 50 Ω.

(b) Define Hamming distance. How is it used in error detection and correction?

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain the construction of a (7, 4) linear block code with its generator matrix.

(b) Define frequency modulation (FM) and phase modulation (PM). Derive their expressions.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Pre-emphasis and de-emphasis in FM

(b) Geostationary satellites & its orbit

(c) Tuned radio frequency (TRF) and superheterodyne receivers

(d) Bandwidth requirement of FM using Carson's Rule

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